AP-E3 UV Follow-up: An Exact Two-Constituent Gauge/Unit-Cell Rule, a Physical $k = +2$ Orientation, and an Anomaly-Consistent Mixed-WZW Completion Candidate

Route F research note

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Abstract

This note closes two sharply delimited AP-E3 questions and exposes the gates that remain. First, two independent compact constraint gauge fields impose $G_r = N_r - 1 = 0$, $r = 1, 2$, on every indivisible unit cell. The physical cell Hilbert space is therefore exactly $\mathbb{C}^2 \otimes \mathbb{C}^2$; singleton, odd-total, and charge-transfer sectors are absent kinematically rather than merely heavy. Ferromagnetic exchange selects the triplet, while the sign of the electron Pauli magnetic moment selects its anti-aligned coherent ground line. If $A_+ = -\mathrm{i}s_+^\dagger \mathrm{d}s_+$ is the AP-E1 unit-flux connection, then

$$\mathrm{i}\hbar\langle\Omega_-|\mathrm{d}\Omega_-\rangle = +2\hbar A_+.$$

Thus the signed worldline level is physically $k = +2$. The ground line is the square of the universal quotient line, $Q^{\otimes 2} \simeq \mathcal{O}(2)$, and directly has three holomorphic sections.

Second, we construct an anomaly-consistent intermediate UV candidate with gauge group $SU(2)_c \times U(1)_g$, two vectorlike Dirac flavours of abelian charge one, and a charge-one scalar doublet. All dynamical local and Witten gauge anomalies cancel. Gauging $U(1)_g$ removes the problematic off-diagonal Pauli–Gürsey generators, and generalized-symmetry matching fixes the mixed-WZW coefficient to $n = n_c X_q = 2$. A positive unit baryon reduces the five-dimensional term to $k = nB = +2$. The same colour Gauss law forces a local baryon to contain two quarks and two conjugate-Higgs dressings, producing a degree-two triplet.

The deterministic audit passes all stated algebraic and numerical checks. Nevertheless, neither branch proves the Route-E family portal. The mixed-WZW model still needs nonperturbative vacuum selection, soliton stability, compact-$U(1)$/bordism tests, and an all-scale completion beyond its abelian Landau pole. Hence the exact constrained-cell theorem and signed orientation are closed within their declared microscopic model, while full AP-E3 UV physics and physics promotion remain false.

Contents

1 Logical target and result ledger

The earlier finite-U Mott–Hund model proves a level magnitude only after declaring two orbitals and an interacting occupancy plateau. Finite energy penalties never remove states from a UV Hilbert space. The first task here is therefore not to increase a Mott gap, but to replace the energetic statement by an exact physical-state condition. The second task is to select the sign of the Berry action from a microscopic coupling rather than relabeling an already derived negative level. The third task is an independent four-dimensional topological mechanism with a complete anomaly ledger.

The conclusions must be read at three distinct levels.

(L1) **Exact constrained-cell theorem.** Once the two compact Gauss laws and complete-cell boundary condition are part of the microscopic definition, exactly two constituents and the signed $k = +2$ ground line are proved.

(L2) **Anomaly-consistent intermediate UV candidate.** The charged two-colour theory cancels every dynamical gauge anomaly and matches the required two-group coefficient, but its strong phase and all-scale UV completion are not proved.

(L3) **Route-E interpretation.** No calculation below identifies the triplet with three chiral $Spin(10)$ families or derives the portal map. Physics promotion therefore remains forbidden.

2 Exactly two as a gauge/physical-state rule

2.1 Why the two Gauss laws are the minimal strong rule

Four superficially similar restrictions are inequivalent.

Rule	Physical sector	Decisive boundary
$G_r = N_r - 1 = 0$, $r = 1, 2$	exactly $(N_1, N_2) = (1, 1)$	removes odd and charge-transfer sectors kinematically; selected here
$G = N_1 + N_2 - 2 = 0$	total number two	still contains $(2, 0)$ and $(0, 2)$; insufficient without another selector
$(-1)^{N_1+N_2} = +1$	every even sector	contains vacuum, $N = 4, 6, \ldots$ and does not select the pair
finite $\Lambda_G \sum_r (N_r - 1)^2$	all sectors, separated in energy	is an EFT approximation and cannot be called an exact rule

The selected cell contains two co-located but internally labelled local spin-1/2 moments. The label may be an orbital, layer, or microscopic dimer label, but a permitted spatial boundary must terminate whole cells. This is a model input: the constraint derives the consequences of the cell, not why nature chose this cell architecture.

2.2 Constraint action, projector, and large gauge invariance

For cell c, internal label $r = 1, 2$, and spinor index $A = 1, 2$, let f_{crA} be *bosonic* Schwinger partons with

$$[f_{crA}, f^\dagger_{c'r'B}] = \delta_{cc'}\delta_{rr'}\delta_{AB}, \qquad N_{cr} = f^\dagger_{crA} f_{crA}, \qquad G_{cr} = N_{cr} - 1. \tag{1}$$

The repeated spinor index is summed. A worldline action that implements the constraint is

$$L_c = \mathrm{i}\hbar \sum_{r,A} f^\dagger_{crA} \dot{f}_{crA} + \hbar \sum_r a_{0,cr}(N_{cr} - 1) - H_c. \tag{2}$$

Under

$$f_{cr} \longmapsto e^{\mathrm{i}\alpha_{cr}} f_{cr}, \qquad a_{0,cr} \longmapsto a_{0,cr} + \dot{\alpha}_{cr}, \tag{3}$$

the Lagrangian changes only by $-\hbar \sum_r \dot{\alpha}_{cr}$. On a time circle, $\Delta\alpha_{cr} = 2\pi m_{cr}$ gives

$$\exp\left(\frac{\mathrm{i}\Delta S}{\hbar}\right) = \exp\left(-2\pi\mathrm{i}\sum_r m_{cr}\right) = 1. \tag{4}$$

Thus the integer background charge one has no $0+1$-dimensional large-gauge obstruction. A half-integer background would fail this test.

The exact physical projector is

$$\Pi^{\mathrm{phys}}_c = \prod_{r=1}^{2} \int_0^{2\pi} \frac{\mathrm{d}\lambda_{cr}}{2\pi} \exp[\mathrm{i}\lambda_{cr}(N_{cr} - 1)]\,. \tag{5}$$

It follows immediately that

$$\mathcal{H}^{\mathrm{phys}}_c = \Pi^{\mathrm{phys}}_c \mathcal{H}_c = \ker G_{c1} \cap \ker G_{c2} \simeq \mathbb{C}^2 \otimes \mathbb{C}^2, \qquad \dim \mathcal{H}^{\mathrm{phys}}_c = 4. \tag{6}$$

All of $(0, 0), (1, 0), (0, 1), (2, 0), (0, 2)$ lie outside this Hilbert space. In particular, exactly two is an operator identity on physical states, not the location of a minimum of a potential.

The gauge-invariant spin operators

$$\boldsymbol{S}_{cr} = \frac{1}{2} f^\dagger_{crA} \boldsymbol{\sigma}_{AB} f_{crB} \tag{7}$$

commute with every $G_{cr'}$. Elitzur's theorem forbids a parton order parameter $\langle f \rangle$, not the physical spin, its spectral projector, or a gauge-invariant Berry holonomy.

2.3 Relation to the old finite-Mott candidate

On the old $(N_1, N_2) = (1, 1)$ sector, the map

$$f_{1A}^\dagger f_{2B}^\dagger |0\rangle_f \longmapsto a_{1A}^\dagger a_{2B}^\dagger |0\rangle_a \tag{8}$$

intertwines both spin algebras. Consequently the Hund spectrum, triplet projector, coherent-state Berry connection, and Veronese map agree exactly. The difference is entirely UV-logical: the old Mott construction retains singleton and doublon states at finite energy, whereas Eq. (1) removes them from the physical Hilbert space. Recent two-band calculations illustrate why a finite-U quasiparticle or local-moment regime must not be confused with an exact projection [1].

2.4 Complete-cell boundary theorem and its limitation

Let $P_{0,c}(\boldsymbol{n}) = |\Omega_{-,c}(\boldsymbol{n})\rangle\langle\Omega_{-,c}(\boldsymbol{n})|$ be the unique local ground projector constructed below and $Q_{0,c} = 1 - P_{0,c}$. If an allowed finite region Λ consists of complete cells and an intercell parent interaction satisfies

$$V_{\text{safe}} \geq 0, \qquad V_{\text{safe}} \bigotimes_{c\in\Lambda} |\Omega_{-,c}\rangle = 0, \qquad [V_{\text{safe}}, G_{cr}] = 0, \tag{9}$$

then the local gap theorem below implies

$$H_\Lambda - E_0|\Lambda| \geq h \sum_{c\in\Lambda} Q_{0,c} + V_{\text{safe}}. \tag{10}$$

The product ground state is unique and $\Delta_\Lambda \geq h$. Thus this safe parent cone has neither a literal cut-cell singleton nor an accidental zero-energy boundary mode.

Fail-closed gate 2.1 (Arbitrary interacting boundaries)**.** The Gauss laws exclude literal odd parton number, but do not forbid every emergent projective edge representation. A sufficiently different intercell interaction may enter a Haldane-like phase and create boundary spin-1/2 modes without violating $N_1 = N_2 = 1$. Outside Eq. (9), a gapped homotopy or a direct boundary-spectrum proof is mandatory. Lieb–Schultz–Mattis-type constraints are a useful warning, not a proof for this particular model [2].

3 A physical orientation and the signed level $k = +2$

3.1 The ferromagnetic sign from exchange

For two localized electronic orbitals φ_1, φ_2, the Coulomb exchange integral can be written

$$K_{12} = \int \frac{\mathrm{d}^3q}{(2\pi)^3} \frac{4\pi e^2}{\epsilon q^2} \left|\widetilde{\varphi_1^*\varphi_2}(\boldsymbol{q})\right|^2 \geq 0. \tag{11}$$

The direct Coulomb contribution is common to singlet and triplet, whereas exchange gives

$$E_{\text{singlet}} - E_{\text{triplet}} = 2K_{12}, \qquad J_H = 2K_{12} > 0. \tag{12}$$

This supplies a first-principles sign for the effective interaction $-J_H \boldsymbol{S}_1 \cdot \boldsymbol{S}_2$, subject to the declared localized-orbital approximation.

3.2 The Pauli-Zeeman sign is the orientation selector

For an electron with charge $-e$ and $g_{\mathrm{eff}} > 0$,

$$\boldsymbol{\mu} = -g_{\mathrm{eff}}\mu_B \boldsymbol{S}, \qquad H_Z = -\boldsymbol{\mu}\cdot\boldsymbol{B} = +g_{\mathrm{eff}}\mu_B B_{\mathrm{mag}}\, \boldsymbol{n}\cdot\boldsymbol{S}, \qquad B_{\mathrm{mag}} = |\boldsymbol{B}|, \qquad \boldsymbol{n} = \frac{\boldsymbol{B}}{B_{\mathrm{mag}}}. \tag{13}$$

Absorbing the units of $\boldsymbol{S}$ into $h = g_{\mathrm{eff}}\mu_B B_{\mathrm{mag}} > 0$, the cell Hamiltonian is

$$H_c(\boldsymbol{n}) = -J_H \boldsymbol{S}_{c1}\cdot\boldsymbol{S}_{c2} + h\boldsymbol{n}\cdot(\boldsymbol{S}_{c1}+\boldsymbol{S}_{c2}), \qquad J_H, h > 0. \tag{14}$$

The plus sign is fixed by the negative electronic charge. It is not a manual reversal of the effective Berry term.

Using $\boldsymbol{S}_1\cdot\boldsymbol{S}_2 = \frac{1}{2}(\boldsymbol{S}_{\mathrm{tot}}^2 - \frac{3}{2})$, the exact spectrum is

$$E_{1,m} = -\frac{J_H}{4} + hm, \qquad m = -1, 0, +1, \tag{15}$$

$$E_{0,0} = \frac{3J_H}{4}. \tag{16}$$

The unique ground state is the anti-aligned triplet $m = -1$, with

$$E_0 = -\frac{J_H}{4} - h, \qquad \Delta_{\mathrm{loc}} = \min\{h, 2h, J_H + h\} = h. \tag{17}$$

Equivalently, as an operator on $\mathcal{H}_c^{\mathrm{phys}}$,

$$H_c - E_0 \geq h\big(1 - P_{0,c}(\boldsymbol{n})\big). \tag{18}$$

3.3 Anti-aligned eigenline as the universal quotient

Using the standard adiabatic eigenline construction [3], let

$$s_+(\theta,\phi) = \begin{pmatrix} \cos(\theta/2) \\ e^{\mathrm{i}\phi}\sin(\theta/2) \end{pmatrix}, \qquad A_+ = -\mathrm{i}s_+^\dagger \mathrm{d}s_+ = \frac{1-\cos\theta}{2}\mathrm{d}\phi. \tag{19}$$

An eigenvector of $\boldsymbol{n}\cdot\boldsymbol{\sigma}$ with eigenvalue -1 is

$$s_- = \mathrm{i}\sigma_y s_+^* = \begin{pmatrix} e^{-\mathrm{i}\phi}\sin(\theta/2) \\ -\cos(\theta/2) \end{pmatrix}, \qquad -\mathrm{i}s_-^\dagger \mathrm{d}s_- = -A_+ \tag{20}$$

up to an exact patch-gauge term.

There is a useful geometric subtlety. Let

$$0 \longrightarrow S = \mathcal{O}(-1) \longrightarrow \underline{\mathbb{C}}^2 \longrightarrow Q \longrightarrow 0 \tag{21}$$

be the universal sequence. The Hermitian orthogonal complement $S^\perp$ is smoothly and complex-linearly isomorphic to the quotient $Q \simeq \mathcal{O}(1)$. The normalized embedded frame $s_- = \mathrm{i}\sigma_y s_+^*$ is anti-holomorphic in the affine coordinate, but that frame does not define the holomorphic structure. Transporting the quotient holomorphic structure gives $S^\perp \simeq Q$.

On the north–south overlap,

$$s_{+,S} = e^{-\mathrm{i}\phi}s_{+,N}, \qquad s_{-,S} = e^{+\mathrm{i}\phi}s_{-,N}, \tag{22}$$

and algebraic quotient frames obey $q_S = -wq_N$. Therefore the pair line is

$$L_- = Q^{\otimes 2} \simeq \mathcal{O}(2). \tag{23}$$

3.4 Berry action, Chern number, and three states

The anti-aligned ground state is

$$|\Omega_-(\boldsymbol{n})\rangle = |s_-\rangle_1 \otimes |s_-\rangle_2. \tag{24}$$

Tensor-product differentiation gives

$$\langle \Omega_- | \mathrm{d}\Omega_- \rangle = 2 s_-^\dagger \mathrm{d}s_-, \qquad \boxed{\mathrm{i}\hbar \langle \Omega_- | \mathrm{d}\Omega_- \rangle = +2\hbar A_+.} \tag{25}$$

Relative to $S_k = \hbar k \int A_+$, this is the signed result

$$\boxed{k = +2.} \tag{26}$$

The action connection $\mathcal{B} = \mathrm{i}\langle \Omega_- | \mathrm{d}\Omega_- \rangle$ has

$$\mathrm{d}\mathcal{B} = 2\mathrm{d}A_+, \qquad \frac{1}{2\pi}\int_{\mathbb{CP}^1} \mathrm{d}\mathcal{B} = 2. \tag{27}$$

Consequently positive-polarization quantization directly gives

$$H^0(\mathbb{CP}^1, L_-) = H^0(\mathbb{CP}^1, \mathcal{O}(2)) = \mathrm{Span}\{z_0^2, z_0 z_1, z_1^2\}, \qquad \dim = 3. \tag{28}$$

This corrects a possible overstatement: the normalized orthogonal eigenvector is anti-holomorphic in w, but its quotient holomorphic line is still $\mathcal{O}(1)$, so the pair is the ordinary positive line $\mathcal{O}(2)$.

Fail-closed gate 3.1 (Portal degree)**.** If Route E supplies an effective orientation map $\widehat{\boldsymbol{B}}_{\mathrm{eff}} : \mathbb{CP}^1_E \to S^2$, then pullback gives

$$k_E = 2\deg(\widehat{\boldsymbol{B}}_{\mathrm{eff}}). \tag{29}$$

The present calculation proves the coefficient two on the microscopic orientation sphere, not $\deg = +1$ for an unknown Route-E portal.

4 An anomaly-consistent mixed-WZW intermediate UV candidate

4.1 Field content

Consider the renormalizable gauge theory

$$G_{\mathrm{gauge}} = SU(2)_c \times U(1)_g \tag{30}$$

with $N_f = 2$ vectorlike Dirac flavours and a scalar doublet. In an all-left Weyl notation the relevant representations are

Field	$SU(2)_c$	$U(1)_g$	global representation
q_L^{ai}	**2**	$+1$	**2** of $SU(2)_L$
$q_R^{c\,ai}$	**2**	-1	$\overline{\mathbf{2}}$ of $SU(2)_R$
ϕ_I	**1**	$+1$	**2** of $SU(2)_\phi$

Here $a = 1, 2$, $i = 1, 2$, and $I = 1, 2$. A potential $V_\phi = \lambda(\phi^\dagger\phi - v^2)^2$ produces

$$\{\phi^\dagger\phi = v^2\}/U(1)_g \simeq S^3/U(1) \simeq \mathbb{CP}^1. \tag{31}$$

The QCD-like condensate is required to break $SU(2)_L \times SU(2)_R \to SU(2)_V$, with pion field $g \in SU(2)$.

4.2 Why pure two-colour QCD is a negative control

Without $U(1)_g$, pseudoreality of the colour fundamental enlarges the flavour group to $SU(2N_f)$ and the standard condensate gives

$$SU(2N_f) \longrightarrow Sp(2N_f). \tag{32}$$

At $N_f = 2$, $SU(4)/Sp(4) \simeq S^5$ and $\pi_3(S^5) = 0$. There is then no stable Skyrmion number to insert into $k = n_c B$. This is why simply substituting $n_c = 2$ into an ordinary QCD formula is invalid. Recent two-colour effective studies explicitly retain the Pauli–Gürsey structure and diquark directions [4].

Gauging $U(1)_g$ changes this symmetry statement exactly. In the Pauli–Gürsey basis $Q = (q_L, \epsilon_c q_R^c)$, the gauge generator is

$$T_g = \mathrm{diag}(\mathbf{1}_{N_f}, -\mathbf{1}_{N_f}). \tag{33}$$

Its centralizer is

$$C_{SU(2N_f)}(T_g) = S\big(U(N_f)_L \times U(N_f)_R\big). \tag{34}$$

Every off-diagonal Pauli–Gürsey generator changes $U(1)_g$ charge by two and is not a global current. This removes the symmetry obstruction, but does not by itself prove which condensate strong dynamics selects.

4.3 Complete dynamical gauge-anomaly ledger

The perturbative cubic $SU(2)_c^3$ anomaly vanishes because the fundamental is pseudoreal. The remaining local gauge-anomaly coefficients are

$$\begin{aligned}
\mathcal{A}_{SU(2)_c^2 U(1)_g} &= N_f T(\mathbf{2})(1-1) = 0, && (35)\\
\mathcal{A}_{SU(2)_c U(1)_g^2} &= 2N_f\big(1^2 + (-1)^2\big)\,\mathrm{Tr}\, T^a = 0, && (36)\\
\mathcal{A}_{U(1)_g^3} &= 2N_f\big(1^3 + (-1)^3\big) = 0, && (37)\\
\mathcal{A}_{\mathrm{grav}^2 U(1)_g} &= 2N_f(1-1) = 0. && (38)
\end{aligned}$$

There are $2N_f = 4$ left-Weyl colour doublets, so the mod-two Witten anomaly also vanishes. Scalars do not contribute to fermionic anomalies. Thus the dynamical gauge theory passes both perturbative and $SU(2)$ global anomaly tests.

Background flavour fields reveal the generalized-symmetry datum rather than a gauge inconsistency. Let b_g be the dynamical $U(1)_g$ connection, $h_g = \mathrm{d}b_g$, and let $F^{\mathrm{fl}}_{L,R}$ be the two background flavour curvatures. Here $n_c = 2$ and the quark charge is $X_q = 1$. In this normalization, the mixed part of the six-form anomaly polynomial contains

$$I_6 \supset \frac{n_c X_q}{2}\frac{h_g}{2\pi}\left[\mathrm{Tr}\left[\left(\frac{F^{\mathrm{fl}}_L}{2\pi}\right)^2\right] - \mathrm{Tr}\left[\left(\frac{F^{\mathrm{fl}}_R}{2\pi}\right)^2\right]\right], \tag{39}$$

so the Postnikov classes are

$$\widehat{\kappa}_L = -\widehat{\kappa}_R = n_c X_q = 2. \tag{40}$$

Equation (39) is written in a Bardeen scheme that preserves the two dynamical gauge symmetries. Its background-flavour variation is compensated by the transformation of the magnetic one-form background; this is the two-group datum, not a residual gauge anomaly. Each background $SU(2)_{L,R}$ also sees two colour copies, an even number, so its Witten anomaly is trivial. The scalar $SU(2)_\phi$ acts only on bosons. Bordism refinements remain necessary for the discrete axial and faithful-group quotients [5, 6].

4.4 Global definition and coefficient of the mixed WZW term

Whenever the physical four-manifold, the sigma-model field, and the relevant bundles extend over a five-manifold Y_5, choose integrally normalized forms

$$\int_{S^3} \omega_3(g) = 1, \qquad \omega_2(s) = \frac{F_s}{2\pi}, \qquad \int_{S^2} \omega_2 = 1, \tag{41}$$

where the trace convention in $\omega_3 = (24\pi^2)^{-1}\,\mathrm{Tr}(g^{-1}\mathrm{d}g)^3$ is fixed by the first equation. The action is

$$S_{\mathrm{mix}} = 2\pi\hbar n \int_{Y_5} \omega_3(g) \wedge \omega_2(s), \qquad n \in \mathbb{Z}. \tag{42}$$

Two such extensions differ by a closed five-manifold. The difference in $S/2\pi\hbar$ is n times an integer, so the exponentiated action is extension-independent. Non-extendible sectors require a differential-character or equivalent Čech/bordism definition; constructing that refinement remains an explicit global-definition gate here. The UV two-group matching fixes, rather than merely permits,

$$\boxed{n = n_c X_q = 2.} \tag{43}$$

This coefficient is quantized and tree-level exact in the construction of Davighi and Lohitsiri [7].

The local sign can also be followed through the tree-level matching. At energies below the Higgs scale, neglecting the b_g kinetic term at leading derivative order gives

$$\begin{aligned} \mathcal{L}[b_g] &= v^2 b_{g,\mu} b_g^\mu + b_{g,\mu}\big(j_\phi^\mu + X_q j_q^\mu\big), \\ b_g^\mu &= -\frac{j_\phi^\mu + X_q j_q^\mu}{2v^2}, \qquad \mathcal{L}_{\mathrm{eff}} = -\frac{(j_\phi + X_q j_q)^2}{4v^2}. \end{aligned} \tag{44}$$

Confinement identifies $j_q = n_c j_B$. Locally define $A_\phi = j_\phi/(2v^2) = +\mathrm{i}s^\dagger \mathrm{d}s = -A_R$, where $A_R = -\mathrm{i}s^\dagger \mathrm{d}s$ is our Route-E/AP-E1 convention and $F_s = \mathrm{d}A_R$. The cross term is therefore

$$S_{\mathrm{cross}} = -n_c X_q \hbar \int *j_B \wedge A_\phi = +n_c X_q \hbar \int *j_B \wedge A_R. \tag{45}$$

Thus the microscopic current coupling fixes the local sign; the five-dimensional expression supplies its global, quantized completion.

4.5 Worldline reduction and physical orientation

For a baryon/Skyrmion with

$$B = \int_{M_3} \omega_3(g) \in \mathbb{Z}, \tag{46}$$

take $Y_5 = M_3 \times D_2$, with $\partial D_2 = \gamma$. Product orientation and Stokes' theorem give

$$\begin{aligned} S_{\mathrm{mix}} &= 2\pi\hbar n \left(\int_{M_3} \omega_3\right)\left(\int_{D_2} \frac{F_s}{2\pi}\right) \\ &= \hbar n B \oint_\gamma A_R. \end{aligned} \tag{47}$$

Therefore

$$\boxed{k = nB = 2B.} \tag{48}$$

The positively oriented baryon sector $B = +1$ has $k = +2$; its CPT antibaryon has $B = -1$ and $k = -2$. This orientation statement is physical but sector-dependent, not a claim of a net CPT asymmetry.

4.6 Colour Gauss law, Higgs dressing, and the even-level rule

One colour fundamental is not a local gauge singlet. The minimal local baryon contains two quarks and carries $U(1)_g$ charge $+2$. A local gauge-invariant interpolating operator must therefore contain two conjugate Higgs fields:

$$\mathcal{B}_{ij}^{IJ} \sim \epsilon_{ab}(q_i^{aT} C q_j^b)\phi^{\dagger(I}\phi^{\dagger J)}. \tag{49}$$

The scalar factor is

$$\mathrm{Sym}^2\,\overline{\mathbf{2}} = \mathbf{3}, \qquad \mathcal{O}(1)^{\otimes 2} = \mathcal{O}(2). \tag{50}$$

The antisymmetric same-point scalar singlet vanishes. Thus the first local colour-singlet baryon is simultaneously an exactly-two scalar dressing, a degree-two line, and an $SU(2)_\phi$ triplet.

There is also a faithful-group explanation of evenness. The complete triple

$$(-\mathbf{1}_c, e_g^{\mathrm{i}\pi}, -\mathbf{1}_\phi)$$

acts trivially on both quarks and scalars. Equivalently, the global centre $-\mathbf{1}_\phi$ agrees on all fields with the gauge transformation $(-\mathbf{1}_c, e_g^{\mathrm{i}\pi})$. Hence the faithful global action on local gauge-invariant scalar operators is

$$SU(2)_\phi/\mathbb{Z}_2 \simeq SO(3). \tag{51}$$

An $SO(3)$-equivariant $\mathcal{O}(k)$ line requires $(-1)^k = 1$. The local physical sector therefore has even k, with minimal nonzero value two. Nonlocal line defects are an explicit exception and may carry projective representations.

5 What remains before this is a full UV completion

5.1 Strong phase and scale ordering

The charged theory removes the exact Pauli–Gürsey symmetry, but a desired vacuum still has to be selected dynamically. The required inequalities are schematically

$$m_{\mathrm{PG}}^2 > 0, \qquad m_{\mathrm{PG}} \gg E_{\mathrm{worldline}}, \qquad \Lambda_c \gtrsim m_b \sim e_g v. \tag{52}$$

The first two prevent the baryon from unwinding through an approximately S^5 diquark direction. The last ordering keeps confinement in a regime where $U(1)_g$ still distinguishes the two Pauli–Gürsey blocks. If $v \gg \Lambda_c$, the heavy vector leaves only suppressed interactions and the unwanted symmetry may become approximately accurate. These questions need lattice or another controlled nonperturbative calculation.

5.2 Running and the all-scale obstruction

With two Dirac flavours, two colours, and two charge-one complex scalars, the one-loop coefficients are

$$b_{U(1)} = \frac{4}{3}(2N_f) + \frac{1}{3}(2) = 6, \qquad b_{SU(2)_c} = \frac{11}{3}(2) - \frac{4}{3}T(\mathbf{2})N_f = 6. \tag{53}$$

Colour is asymptotically free, but the abelian coupling has a Landau pole. The theory is therefore a valid anomaly-consistent intermediate UV description above the sigma model, not an all-scale completion.

A proposed nonabelian embedding $SU(3) \to [SU(2)_c \times U(1)]/\mathbb{Z}_2$ has a sharp representation-theoretic cost: every $SU(2)_c$ singlet descending from an $SU(3)$ representation has even abelian charge. A light colour-singlet charge-one ϕ_I must therefore arrive with coloured partners, whose decoupling must preserve both light scalars and $SU(2)_\phi$. No such complete decoupling proof is provided here.

One creative research branch replaces $U(1)_g$ by an asymptotically free $SU(2)_g$, broken by an adjoint to its Cartan. Bifundamental quarks and scalar $SU(2)_g$ doublets could reproduce the intermediate spectrum, but mass splitting makes the chiral symmetry emergent and 't Hooft–Polyakov monopoles make the magnetic one-form symmetry only approximate. It is a future branch, not a claimed solution.

5.3 Soliton and quantization gates

Even a quantized $k = 2$ term does not prove a three-state low-energy particle. The following remain independent calculations:

1. existence and metastability of a $B = 1$ soliton in the full field space, including its Hessian and decay channels;
2. Finkelstein–Rubinstein statistics and the collective-coordinate moment of inertia;
3. separation of the $\mathcal{O}(2)$ lowest sector from the higher rotor tower;
4. compact-$U(1)$ monopole breaking of the magnetic one-form symmetry;
5. discrete axial and spin-bordism consistency;
6. a Route-E portal whose degree is $+1$ and whose couplings remain below every retained gap.

6 Deterministic regression and numerical evidence

The companion verifier `route_f/code/verify_ap_e3_exact_two_mixed_wzw.py` performs the following independent checks.

1. It enumerates the bosonic four-mode Fock basis truncated at $0 \leq N_r \leq 2$ and evaluates the exact Fourier projector (5). The separate projector has rank four; within this declared audit truncation, the single total-number projector has rank ten and retains both charge-transfer sectors. The exact constrained rank four is truncation-independent.
2. It diagonalizes (14) at $J_H = 1, h = 0.2$. The spectrum is

$$(-0.45, -0.25, -0.05, 0.75), \tag{54}$$

   and seeded random directions agree with the anti-aligned tensor-square projector to floating-point precision.
3. Exact directional derivatives verify $\mathrm{i}\langle\Omega_-|\mathrm{d}\Omega_-\rangle = 2A_+$ and doubled quantum metric. Midpoint curvature quadrature converges monotonically to $c_1 = +2$, while the overlap transition has winding two.
4. The script evaluates every local anomaly sum, the mod-two colour and flavour counts, $\widehat{\kappa}_L = -\widehat{\kappa}_R = 2$, extension independence of the exponentiated five-dimensional action, and $k(B = +1) = +2$.
5. Negative controls include one total Gauss law, parity-only projection, a half-integer background charge, pure two-colour QCD, and the abelian Landau pole.

These tests validate formulas and conventions; they cannot replace the nonperturbative phase and soliton calculations listed above.

7 Claim ledger and ordered continuation

Claim	Status	Boundary
Separate compact Gauss laws impose $N_1 = N_2 = 1$	proved	within the declared indivisible cell
Literal odd and charge-transfer sectors are absent	proved	not a finite-energy statement
Safe-parent complete-cell boundaries have gap at least h	proved	arbitrary intercell phases excluded
Pauli orientation gives signed $k = +2$	proved	on the microscopic orientation sphere
Positive ground line is $Q^2 = \mathcal{O}(2)$ with three sections	proved	family interpretation not implied
Dynamical gauge anomalies of the charged two-colour model cancel	proved	discrete/bordism refinements remain
Mixed-WZW coefficient and $B = 1$ reduction give $k = +2$	proved conditionally	extendible sectors; assumes the desired gapped strong phase and unit soliton
Global differential-cohomology definition on non-extendible sectors	open	extension independence alone is insufficient
All-scale mixed-WZW UV completion	open	abelian Landau pole and nonabelian decoupling problem
Route-E degree-one chiral-family portal	open	no microscopic map or anomaly-complete family operator

The next steps should be executed in this order:

1. lattice or another nonperturbative audit of the charged two-colour phase, measuring meson/diquark condensates, $m_{\rm PG}$, and the $B = 1$ lifetime over $(e_g, v/\Lambda_c)$;
2. APS/bordism and compact-monopole audit of the faithful global group;
3. collective-coordinate Hessian, statistics, and rotor/LLL gap;
4. explicit degree-one Route-E portal below all gaps;
5. use the now-complete AP-E4 mathematical spectrum as an independent control, then derive its still-open physical fermion origin through either moduli-space SQM or an anomaly-complete product compactification.

The fail-closed status is therefore

$$\boxed{\begin{gathered}\texttt{exactly\_two\_constrained\_cell\_proved=true},\\ \texttt{physical\_pauli\_orientation\_gives\_k\_plus\_2=true},\\ \texttt{dynamical\_gauge\_anomalies\_cancel=true},\\ \texttt{mixed\_wzw\_intermediate\_uv\_candidate=true},\\ \texttt{mixed\_wzw\_full\_uv\_closed=false},\\ \texttt{route\_e\_portal\_closed=false},\\ \texttt{physics\_promotion\_allowed=false}.\end{gathered}} \tag{55}$$

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